NASH ALTHEA G. TOLIN / RESEARCH HIGHLIGHT

FINGERPRINT-BASED ATTENDANCE SYSTEM

Senior high school capstone | Academic Year 2025–2026

Project overview

A collaborative research project evaluating a fingerprint-based school attendance monitoring system with a kiosk and web access for teachers, parents and administrators. Nash Althea G. Tolin is one of eight named researchers in the original study.

System components

ESP32 microcontroller and fingerprint sensor • Arduino IDE

Kiosk for enrollment and student time-in/time-out

Google Sheets and Google Apps Script for attendance records and web access

Research approach

Descriptive quantitative design using surveys and an observation checklist.
Participants: 39 Grade 12 STEM students, 3 teachers and 10 parents/guardians.

Selected observed results

75% — Correct recognition of enrolled fingerprints on the first attempt (6 of 8 trials).

100% — Rejection of unregistered fingerprints (8 of 8 observed trials).

100% — Correct time-in/time-out recording and automatic transmission (8 of 8).

The study also reports usability and device compatibility issues to improve.

Note: Percentages describe the study's reported tests, not universal system guarantees.
This public highlight intentionally excludes the original document's private contact details.

SOURCE: 2025–2026 STUDENT CAPSTONE RESEARCH (CURATED PUBLIC SUMMARY)